When Blind Sampling Beats Tuned NSGA-II: A Two-Condition Diagnostic with a Calibrated Visa Allocation Case Study

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Abstract. Many hard-constrained combinatorial problems are solved by pairing a metaheuristic with a feasibility-preserving decoder, yet the true driver of performance is seldom isolated. We isolate it on a calibrated case study: allocating roughly 140,000 U.S. employment-based visas per year under a per-country ceiling and a spillover cascade, cast as a three-objective integer program under six hard constraints and solved with multi-objective Harris Hawks Optimization (MOHHO) through a smallest-position-value random-key mapping composed with a greedy assignment feasible by construction—no penalties, no repair. A controlled nine-method ladder shows why search fails here: blind random sampling beats a tuned real-coded NSGA-II and a naive swarm because their realized search is *degenerate*—interpolating operators barely change the decoded order, or gated acceptance freezes the population. These two conditions are necessary but not sufficient: a BRKGA-style biased-uniform crossover that does change the order under diversity-preserving selection only *ties* blind sampling, and performance rises monotonically with order disruption, so that only near-complete order renewal—a competent random-key swarm or permutation-native operators across four paradigms—reaches the top tier. We package these measurements as a cheap a priori diagnostic protocol. In replications on a knapsack, a traveling salesman, and a flow-shop, a method satisfying both conditions is the best optimizer on every structure, while the collapse of the tuned genetic algorithm below blind sampling is specific to *selection* landscapes. Parameters are fixed by a Taguchi design; the recovered front Pareto-dominates the incumbent first-in, first-out policy, yielding an auditable menu of trade-offs.

Keywords: Multi-objective optimization · Harris Hawks Optimization · Constraint handling · Random-key decoder · Permutation encoding · Taguchi design of experiments · Resource allocation

1 Introduction

Each fiscal year the United States Congress authorizes approximately 140,000 employment-based (EB) immigrant visas, distributed across five categories (EB-1 to EB-5). Two structural rules govern the distribution: a *per-country ceiling* ($P_c \approx$ 25,620 visas, the statutory 7 % cross-preference limit over the combined family-sponsored and employment-based preference totals [23]) that no single country may exceed, and a *spillover* mechanism that cascades unused visas across categories (EB-4/5 → EB-1 → EB-2 → EB-3). The incumbent system processes petitions strictly first-in, first-out (FIFO) by priority date. In our calibrated instance, the interaction of these rules creates measurable inefficiencies: two countries concentrate roughly three-quarters of demand (India ≈ 63 %, China ≈ 14 %) and accumulate waits of 10–13 years, while the clash between per-country and per-category ceilings leaves thousands of visas unused.

This matters beyond accounting. Behind every backlogged petition is a household whose relocation, employment, and family reunification hinge on a priority date, so the allocation rule has direct welfare consequences. The problem admits three legitimate, mutually conflicting goals: serve the longest-waiting petitions (*efficiency*), distribute opportunity evenly across nationalities (*equity*), and waste no visas (*utilization*). No single allocation optimizes all three; the policy-relevant object is the set of best compromises—a Pareto front. Algorithmic assignment of people to scarce slots has been studied for refugee resettlement [2]; our contribution is methodological, not a new matching design.

Computing that front is hard for three reasons. First, the allocation is *combinatorial*: it assigns integer visa counts to 105 country×category groups under six simultaneous constraints, and its feasible set generalizes the multidimensional knapsack problem, which is NP-hard [13]. Second, the objectives are genuinely *conflicting*, so the solution is a front rather than a point. Third, the *hard constraints* are unforgiving: a continuous metaheuristic naturally proposes infeasible candidates, and penalties or aggressive repair either require delicate tuning or distort the search landscape.

We address these challenges with a multi-objective adaptation of Harris Hawks Optimization (MOHHO) whose centerpiece is a *feasibility-preserving decoder*. Our central contribution is methodological: a controlled, mechanistically explained account—developed on the visa problem and stress-tested across four structurally distinct multi-objective problems—of *when* decoder-based search beats blind sampling, distilled into a two-condition diagnostic that a practitioner can run before committing a full experimental budget. Concretely, this paper makes five contributions: *(i)* a decoder that couples a smallest-position-value (SPV) random-key mapping with a deterministic greedy assignment, guaranteeing all six policy constraints *by construction*—without penalties or repair, and therefore without perturbing the search landscape; *(ii)* a calibrated, constraint-heavy allocation case study on which a multi-objective Harris Hawks optimizer with a crowding-pruned external archive recovers a policy-relevant Pareto front that dominates the FIFO incumbent; *(iii)* a Taguchi L9 design that fixes the population size and iteration budget shared by *every* ladder method, so that cross-method comparisons are not confounded by per-method tuning; *(iv)* a controlled nine-method ladder (with *Discrete-MOHHO* as swarm-side mechanism check), a 2×2 isolation experiment, and a direct operator-order-preservation measurement (Kendall τ through the decoder) that together yield the two-condition diagnostic—operators must change the decoded order *and* selection must preserve diversity—validated *predictively* on a BRKGA-style crossover not used to formulate it, which sharpens the two conditions from a binary rule to necessary-but-graded; and *(v)* a reproducible empirical protocol (30 seeds, omnibus Friedman/Nemenyi tests with effect sizes, FIFO domination, five perturbed-demand instances) and a four-structure generalization (knapsack, traveling salesman, flow-shop) separating what transfers—a method satisfying both conditions wins on every structure, the family second-order among them—from what is landscape-specific (the GA-below-random collapse).

2 Background and Related Work

Harris Hawks Optimization (HHO) [15] is a population-based metaheuristic whose prey *escape energy* E drives a continuous exploration–exploitation transition, with optional Lévy-flight dives. It has been extended to multiple objectives via external archives, elitist non-dominated sorting, and grid indices [27,26,16,6]; these variants are validated mostly on continuous benchmarks, leaving their use on *constrained combinatorial* allocation—where feasibility, not convergence, is the binding difficulty—comparatively unexplored.

We bridge continuous search and combinatorial structure with random keys [3]: a real vector is decoded to a feasible permutation by sorting (the smallest-position-value, SPV, rule). How much an operator step changes the decoded phenotype is the classical representation-*locality* question, formalized by Rothlauf [21] and measured for decoder-based EAs by Raidl and Gottlieb [20]—the lens our Kendall-τ measurement instantiates for random keys under multi-objective search. The idea underlies biased random-key genetic algorithms (BRKGA) [14,17], whose uniform crossover was designed precisely because interpolating operators move the keys without moving the decoded order, and a general random-key optimizer that pairs a decoder with *any* metaheuristic [5]—the paradigm we instantiate for HHO and probe by varying the metaheuristic while holding the decoder fixed. Among constraint-handling families [19], penalty methods need problem-specific weights and blur feasibility, repair can collapse diversity, and decoder (feasibility-preserving) approaches guarantee feasibility by construction; we adopt the third. NSGA-II (Non-dominated Sorting Genetic Algorithm II) [9] is our benchmark and the standard tool for hard allocation and scheduling [29]; our finding that the family is second-order once operators match the representation speaks to the metaphor-centered critique of metaheuristics [22]. We tune parameters with a Taguchi orthogonal array [18,1] and assess fronts with the hypervolume [31], inverted generational distance and spacing [7], nonparametric tests [11], and the A_{12} effect size [25].

3 Problem Formulation

Decision variables. The problem is discretized into $G = C \times |\mathcal{J}| = 21 \times 5 = 105$ *groups*, where each group g is a (country c, category j) pair. The decision vector is $\mathbf{x} = (x_1, \ldots, x_{105}),\ x_g \in \mathbb{Z}_0^+$, where x_g is the number of visas assigned to group g. Each group has a demand (backlog) n_g and a *waiting weight* $w_g = t_{\text{now}} - d_g$, the years elapsed since its priority date d_g.

Objectives. We minimize three conflicting objectives:

$$f_1(\mathbf{x}) = \frac{\sum_g (n_g - x_g)\, w_g}{\sum_g n_g} \quad \text{unserved waiting load (efficiency),} \tag{1}$$

$$f_2(\mathbf{x}) = \max_{c_1, c_2 \in \mathcal{C}} \left|\bar{W}_{c_1} - \bar{W}_{c_2}\right| \quad \text{maximum inter-country disparity (equity),} \tag{2}$$

$$f_3(\mathbf{x}) = V - \textstyle\sum_g x_g \quad \text{wasted (unassigned) visas (utilization),} \tag{3}$$

where V is the total visa supply and $\bar{W}_c = \big(\sum_{g:\, c(g)=c} x_g w_g\big) / \big(\sum_{g:\, c(g)=c} x_g\big)$ is the visa-weighted mean wait of country c. Thus f_2 measures the gap between the longest- and shortest-waiting nationalities: the smaller f_2, the more equitable the distribution. (A country receiving no visas takes its worst backlog wait as $\bar{W}_c$, so f_2 implicitly penalizes starving a nationality; FIFO and all reported equity solutions serve every country.)

Constraints. The feasible set $\mathcal{F}$ is defined by six hard constraints:

R1: $\sum_g x_g \le V = 140{,}000$	(total visas)
R2: $\sum_{g:\, c(g)=c} x_g \le P_c = 25{,}620\ \ \forall c$	(cross-preference country ceiling)
R3: $\sum_{g:\, j(g)=j} x_g \le K_j^{\text{eff}}\ \ \forall j$	(effective per-category cap)
R4: $0 \le x_g \le n_g\ \ \forall g$	(do not exceed demand)
R5: $x_g \in \mathbb{Z}_0^+\ \ \forall g$	(integrality)
R6: FIFO order within each (c, j) queue	(seniority within a group)

where K_j^{eff} is the per-category cap after the spillover cascade. We use the statutory cross-preference country ceiling as an exogenous annual EB-side cap in this single-year model; a full legal simulation of family-sponsored interactions is outside scope. The cascade is retained for generality but inert here: every category is oversubscribed, so the slack is zero and $K_j^{\text{eff}} = K_j^{\text{base}}$. The result is a three-objective integer program with six constraints whose feasible space grows combinatorially with G, motivating a metaheuristic rather than exact enumeration.

Pareto optimality. A solution $\mathbf{x}$ *dominates* $\mathbf{x}'$ ($\mathbf{x} \prec \mathbf{x}'$) iff $f_i(\mathbf{x}) \le f_i(\mathbf{x}')$ for all i and $f_i(\mathbf{x}) < f_i(\mathbf{x}')$ for some i. The *Pareto front* is $\mathcal{P}^* = \{\mathbf{x} \in \mathcal{F} \mid \nexists\, \mathbf{x}' \in \mathcal{F} : \mathbf{x}' \prec \mathbf{x}\}$. MOHHO approximates $\mathcal{P}^*$ through an external archive.

Table 1. The six MOHHO movement operators, selected by the escape energy $|E|$ and uniform draws q, r. The archive's leader supplies the prey position.

Op.	**Condition**	**Behavior**
OP1	$\lvert E\rvert \geq 1,\ q \geq 0.5$	Exploration from a random hawk in the swarm.
OP2	$\lvert E\rvert \geq 1,\ q < 0.5$	Exploration relative to the population centroid.
OP3	$0.5 \leq \lvert E\rvert < 1,\ r \geq 0.5$	Soft besiege: gradual approach to the prey.
OP4	$\lvert E\rvert < 0.5,\ r \geq 0.5$	Hard besiege: aggressive contraction.
OP5	$0.5 \leq \lvert E\rvert < 1,\ r < 0.5$	Soft besiege with rapid Lévy dives.
OP6	$\lvert E\rvert < 0.5,\ r < 0.5$	Hard besiege with rapid Lévy dives.

4 Methods

4.1 Multi-objective Harris Hawks Optimization

Each hawk is a position $\mathbf{H} \in \mathbb{R}^{105}$ in the box $[0, 1]^{105}$. At iteration t the escape energy is

$$E = 2E_0\left(1 - \tfrac{t}{T}\right), \qquad E_0 \sim U(-1, 1), \tag{4}$$

which drives the choice among six movement operators selected by $|E|$ and uniform draws $q, r \in U(0, 1)$ (Table 1). The multi-objective adaptation draws the *leader* ("rabbit") from an external archive of non-dominated solutions with probability proportional to crowding distance (biasing search toward sparse regions of the front), prunes the most crowded solution on overflow, and updates the archive after every move; the archive size is a tunable parameter (Section 5).

For a function-evaluation-fair comparison, the Lévy operators OP5/OP6 accept a single trial point per hawk per iteration (the canonical HHO evaluates two), so MOHHO consumes exactly $N \times T$ evaluations—identical to every other method.

4.2 A feasibility-preserving decoder

The HHO operators act in the continuous space $\mathbb{R}^{105}$, but the problem is combinatorial with six hard constraints. We bridge the two with the SPV rule [3] followed by a deterministic greedy assignment (Algorithm 1). The SPV rule turns a hawk $\mathbf{H}$ into a permutation $\boldsymbol{\pi}$ by sorting the group indices in ascending order of their key h_g. The greedy decoder then walks the groups in that order and assigns

$$x_g = \min\!\left(n_g,\ V_{\text{rem}},\ P^{\text{rem}}_{c(g)},\ K^{\text{rem}}_{j(g)}\right), \tag{5}$$

decrementing the residual budgets of R1–R3 after each step. Because every assignment is capped by the remaining budgets and by demand, and counts are integers, *every* decoded vector satisfies R1–R6 by construction. No penalty term is introduced and no repair is performed, so the hawks search directly on a feasible objective landscape, preserving HHO's cooperative foraging and stochasticity. The greedy fill is budget-saturating (each visited group gets its

Algorithm 1 Feasibility-preserving SPV + greedy decoder

Require: hawk $\mathbf{H} \in \mathbb{R}^{105}$; demands n_g; budgets V, $\{P_c\}$, $\{K_j^{\text{eff}}\}$
$\boldsymbol{\pi} \leftarrow \textsc{ArgSortAscending}(\mathbf{H})$ ▷ SPV: keys → permutation
$V_{\text{rem}} \leftarrow V$; $P_c^{\text{rem}} \leftarrow P_c\ \forall c$; $K_j^{\text{rem}} \leftarrow K_j^{\text{eff}}\ \forall j$
for $g \in \boldsymbol{\pi}$ **do** ▷ visit groups in SPV order
  $x_g \leftarrow \min\big(n_g, V_{\text{rem}}, P_{c(g)}^{\text{rem}}, K_{j(g)}^{\text{rem}}\big)$
  $V_{\text{rem}} -= x_g$; $P_{c(g)}^{\text{rem}} -= x_g$; $K_{j(g)}^{\text{rem}} -= x_g$
end for
return x ▷ satisfies R1–R6 without repair

maximal feasible amount), so the decoder's image is the saturating subset of $\mathcal{F}$ and all dominance claims are relative to it; an exact bi-objective (f_1, f_3) mixed-integer linear program (MILP) (Section 6.3) confirms the few non-saturating non-dominated allocations that exist improve no objective meaningfully, so this subset loses nothing relevant.

Proposition 1 (Feasibility by construction). *For any hawk* $\mathbf{H} \in \mathbb{R}^{105}$*, the allocation* $\mathbf{x}$ *returned by Algorithm 1 satisfies all six constraints R1–R6.*

Proof. Each step takes $x_g = \min(n_g, V_{\text{rem}}, P_{c(g)}^{\text{rem}}, K_{j(g)}^{\text{rem}})$, so by induction every residual stays ≥ 0 and the caps R1–R3 hold, while $0 \leq x_g \leq n_g$ gives R4 and integrality gives R5. Each group is a single (c, j) pair sharing one priority date, so serving its x_g most-senior petitions honors R6. Thus $\mathbf{x} \in \mathcal{F}$ for every input, with no penalty term and no repair. □

4.3 Benchmarks and a representation-matched optimizer

To separate the contribution of the decoder, the representation, and the search heuristic, we compare a *ladder* of methods that all share the same greedy decoder, objectives, hypervolume reference point, and 50×500 budget; they differ only in how they explore. Two operate on the continuous SPV random keys in $\mathbb{R}^{105}$: *(a)* the MOHHO of Section 4.1; and *(b)* **NSGA-II** [9] with simulated binary crossover (SBX, $\eta_c = 20$, $p_c = 0.9$), polynomial mutation ($\eta_m = 20$, $p_m = 1/d$), binary tournament selection, and elitist $(\mu+\lambda)$ replacement. A third, *(c)* **random restart**, draws the 25,000 candidates uniformly at random (no search), isolating what the decoder alone achieves.

The ablation (Section 6.3) reveals that the continuous random-key operators are a poor match for the induced permutation landscape. We therefore also evaluate **permutation-native** methods—spanning four distinct paradigms—that act directly on permutations of the 105 groups (decoded by the same greedy procedure): *(d)* **permutation-NSGA-II** (dominance-based) with order crossover (OX) and swap mutation; *(e)* **permutation-MOEA/D** (multi-objective evolutionary algorithm based on decomposition [28]; Tchebycheff over 45 simplex-lattice weight vectors, 10-nearest-neighborhood mating and replacement, archive 100, same OX + swap); *(f)* **permutation-SPEA2** [30] (strength-Pareto, density-based environmental selection with truncation) under the same operators; and *(g)* **Discrete-MOHHO**, our representation-matched Harris Hawks optimizer. Discrete-MOHHO keeps HHO's signature—the escape energy E of Eq. (4) scheduling exploration and exploitation, and a crowding-pruned external archive supplying the leader—but expresses every move on permutations: when $|E| \geq 1$ a hawk explores by recombining (OX) with a random hawk or by a heavy random shuffle; when $|E| < 1$ it besieges the leader by inheriting an order segment from it (OX) and perturbing with a number of swaps that shrinks as $|E| \to 0$, with occasional Lévy-like segment reversals. Discrete HHO variants exist for single-objective sequencing; Discrete-MOHHO here is the matched swarm the mechanism check requires, not a novelty claim. Finally we add *(h)* **rk-NSGA-II (biased uniform)** [14]: the skeleton of *(b)* with SBX and polynomial mutation replaced by BRKGA-style parameterized uniform crossover (elite-parent gene w.p. 0.7) and per-gene random-reset mutation—an order-changing random-key control used as a predictive test in Section 6.3. All methods run for 30 independent seeds (1–30) under the identical budget.

4.4 Evaluation protocol and indicators

We assess each approximation front with four indicators: the *hypervolume* (HV; larger is better) dominated with respect to the reference point (10; 16; 20,000), an upper bound on each objective; the *inverted generational distance* (IGD; smaller is better) to a reference front Z formed by the non-dominated union of the MOHHO and NSGA-II combined fronts ($|Z| = 126$; used only for that pairwise comparison—the ladder relies on HV); the *spacing* indicator (smaller is better) measuring uniformity; and the *cardinality* of the non-dominated set. Objectives are min–max normalized before IGD and spacing. Each algorithm is run for 30 independent seeds, and a *combined front* is formed as the non-dominated union of its 30 per-run fronts. Statistical significance between the two 30-value HV distributions is assessed with a one-sided Mann–Whitney U test [11], and effect size with the Vargha–Delaney A_{12} [25]. The deterministic **FIFO** baseline orders groups by ascending priority date and is decoded with the same greedy procedure, yielding $f_1 = 8.7891$, $f_2 = 13.0$ years, and $f_3 = 1{,}940$ wasted visas.

5 Parameter Tuning by Taguchi Orthogonal Array

We replace empirical parameter choices by a Taguchi L9(3^4) orthogonal-array design [18,1] over four factors at three levels—population $N \in \{30, 50, 70\}$, iteration budget $T \in \{100, 300, 500\}$, archive size $\in \{50, 100, 150\}$, and Lévy exponent $\beta \in \{1.3, 1.5, 1.7\}$—with the larger-the-better signal-to-noise (S/N) ratio of the hypervolume as response, each of the nine configurations replicated over 12 seeds disjoint from the evaluation seeds.

Averaging the S/N ratio per factor level (Table 2, Figure 1) gives an unambiguous ranking: *population* ($\Delta = 0.318$ dB) and *iteration budget* ($\Delta = 0.155$ dB) dominate, while β (0.108) and archive size (0.046) lie within run-to-run noise;

Table 2. Response table: mean S/N (dB) per factor level, range Δ, and rank. Larger Δ means a more influential factor; the highest-S/N level in each row is in bold. Each Δ is computed from unrounded S/N values.

Factor	Level 1	Level 2	Level 3	Δ	Rank
A: population N	109.314	109.443	**109.632**	0.318	1
B: iteration budget T	109.372	109.489	**109.528**	0.155	2
C: archive size	109.461	**109.487**	109.441	0.046	4
D: Lévy exponent β	109.462	**109.518**	109.410	0.108	3

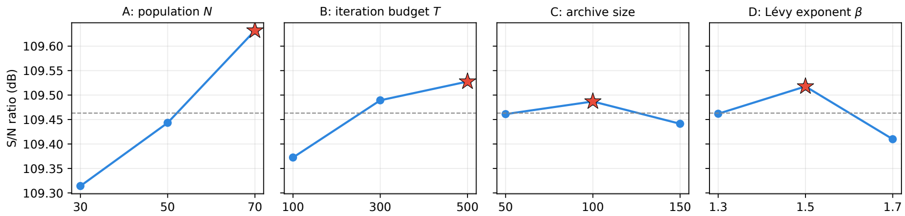

Fig. 1. Main-effects plot of the S/N ratio. The dashed line is the grand mean (109.46 dB) and the red star the best level per factor; population and iteration budget show clear trends (population the steeper), while archive size and β are essentially flat.

the population effect grows toward the top of its range while the iteration effect saturates past T=300. The additive model predicts the optimum N=70, T=500 at S/N $\approx$ 109.77 dB; a confirmation run gives 109.66 dB (HV = 304,391), validating the model. The 30-run study adopts N=50, T=500, archive 100, β=1.5, whose confirmation gives S/N 109.56 dB—within 1.2 % hypervolume of the L9 optimum (its sole difference is N=50 vs. 70), a deliberately conservative, compute-economical operating point (T=500 far exceeds the iteration-135 convergence point).

6 Results

6.1 Combined Pareto front and domination of FIFO

Combining and re-filtering the 30 runs yields a front of **104 non-dominated solutions**. Figure 2 shows it in the full objective space and Figure 3 in the f_1–f_2 projection; in both, the FIFO baseline floats *above* the front, visual evidence that it is dominated. The extreme solutions (Table 3) expose the trade-offs.

The central empirical finding is that **FIFO is not Pareto-optimal**: the minimum-f_1 solution Pareto-dominates it—strictly better on waste (f_3 from 1,940 to 680 visas), tied on the maximal disparity ($f_2 = 13.0$), and only negligibly better on the near-degenerate f_1 (8.7891 to 8.7884). The substantive case against FIFO is thus about waste and equity: 94 of the 104 front solutions reach $f_3 = 0$

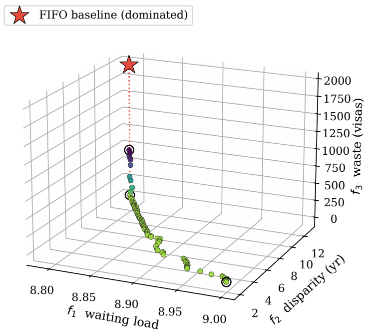

Fig. 2. The 104-solution combined front, colored by waste f_3. The FIFO baseline (red star, isolated above the surface; dotted drop-line) is dominated; circles mark the extreme solutions.

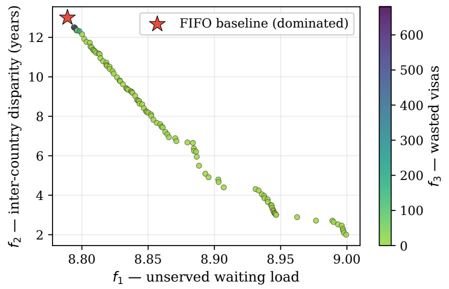

Fig. 3. f_1–f_2 projection of the same front, colored by f_3 (wasted visas). FIFO lies inside the dominated region.

Table 3. Extreme solutions of the combined front, compared with the FIFO baseline. The minimum-f_1 solution dominates FIFO (better on two objectives, equal on the already-maximal disparity).

Solution	f_1	f_2 **(yr)**	f_3 **(visas)**	**Remark**
Min. f_1 (least waiting)	8.7884	13.0	680	**Dominates FIFO**
Min. f_2 (most equitable)	8.9994	2.0	0	Radical equity
Min. f_3 (full use)	8.996	2.46	0	100 % utilization
FIFO baseline	8.7891	13.0	1,940	Reference (dominated)

(zero waste, which FIFO never attains), with disparities spanning from 13.0 years down to 2.0. (Top-tier combined fronts agree within 0.3 % in hypervolume, so the policy menu does not hinge on the optimizer choice.)

6.2 Convergence and robustness

The mean hypervolume reaches 95 % of its final value by iteration 5 and 99 % by iteration 135; gains beyond iteration 140 are under 1 %. Over the 30 runs the HV distribution is narrow (mean 302,756, coefficient of variation [CV] 2.36 %), evidencing a stable, reproducible algorithm.

6.3 Decoder, representation, and search: a controlled ladder

We attribute performance to its true source by climbing a ladder of methods, all sharing the decoder and a 25,000 function-evaluation budget—the comparison currency (Table 4, Figure 4); the permutation-native methods inherit the Taguchi-tuned N and T and are not separately tuned, so any edge is not a tuning artifact.

Under this budget MOHHO beats the real-coded NSGA-II by 3.2 % in mean hypervolume (one-sided Mann–Whitney $p \approx 1.8 \times 10^{-6}$, $A_{12} = 0.85$) and with lower spacing (NSGA-II edges it on IGD to the combined reference front).

Blind sampling beats near-identity search—but not a competent swarm. Random restart—blind sampling with no search—already reaches a mean HV of 310,214: 5.7 % *above* the real-coded NSGA-II and 2.5 % *above* the classic real-coded MOHHO (the naive swarm wins on only 10/30 paired seeds). Under an identical budget the decoder shapes a landscape on which blind sampling outperforms both of these real-coded searches; giving NSGA-II the *identical* archive barely helps it (293,666 vs. 293,367), so its deficit is not an archiving artifact. We are careful, however, not to over-read this as "the decoder does the work": blind sampling beats these two searches because their realized search is degenerate, not because real-coded search is hopeless. To show this, we built a *competent* real-coded MO-HHO—HHO movement with elitist non-dominated-sorting selection and polynomial mutation in place of the dominance-gated acceptance—which is sound where the naive swarm collapses (ZDT2 hypervolume 0.99 vs. 0.20 of the true front) and, at the same 25,000-evaluation budget on the visa, *beats* random restart (316,347 vs. 310,214, +2.0 %, paired Wilcoxon $p = 5.7 \times 10^{-4}$, 30 seeds). So a real-coded swarm *can* beat blind sampling; the naive one does not.

Why near-identity operators fail. Why does a tuned NSGA-II fall below blind sampling? We measured the cause directly. On the SPV random keys, SBX and polynomial mutation are *near-identity on the decoded permutation*: over 3,000 trials the Kendall rank correlation between a parent's SPV order and its child's is $\tau = 0.99$ (and stays in $[0.987, 0.992]$ when measured on-trajectory). Sweeping the crossover spread $\eta_c \in \{2, \dots, 100\}$ confirms this is no tuning artifact: even the most disruptive setting (η_c=2, τ=0.92) leaves the GA's mean HV at 305,293, below blind random restart, and τ and HV move in strictly opposite directions across the sweep. The GA perturbs the keys so little that the induced permutation barely changes, leaving it almost frozen in combinatorial space—precisely why it sinks below blind sampling, which at least draws fresh permutations. HHO's position updates, by contrast, *propose* large permutation jumps ($\tau \approx 0$) that the dominance-gated acceptance rarely admits (only ∼0.6 % of hawks move per iteration), but the archive keeps the rejected proposals—hence MOHHO > NSGA-II within the random-key tier. The real-coded MOHHO is thus a deliberately standard, *weak* baseline (on ZDT2 its operators collapse the population to a corner, ∼0.20 of the true-front hypervolume under every acceptance rule and repair we tried): our thesis rests on degenerate real-coded search losing even to blind sampling, not on the swarm being strong.

Matching the operators to the representation. The clean fix is to act *directly* on permutations. Doing so lifts *four* distinct paradigms above blind sampling: permutation-NSGA-II (dominance/GA) reaches 318,151, permutation-SPEA2 (strength-Pareto) 317,135, Discrete-MOHHO (swarm) 316,792, and permutation-MOEA/D (decomposition) 314,846—and the competent random-key swarm joins them (316,347), so the encoding is not the divider. The four cluster within about 1 % of one another (permutation-SPEA2 vs. permutation-NSGA-II: two-sided Wilcoxon $p = 0.47$, $A_{12} = 0.45$, a statistical tie), yet all sit 1.5–2.6 % above blind random restart—more than the spread *among* them. Discrete-MOHHO improves on classic MOHHO by +4.6 % (paired Wilcoxon $p = 9.3 \times 10^{-10}$, better on all 30/30 seeds); the most stable methods are matched ones (permutation-NSGA-II CV 0.58 %, Discrete-MOHHO 0.71 %, vs. classic MOHHO 2.36 %), with permutation-NSGA-II holding the marginal mean lead (+0.43 % over Discrete-MOHHO). Discrete-MOHHO is therefore not the overall champion but a mechanism check: made representation-matched, the swarm reclaims the top tier and low variance while permutation-NSGA-II remains marginally strongest overall.

The two conditions, graded by a predictive test. The ladder suggests a two-condition rule for this landscape: a method beats blind sampling only if its operator changes the decoded order (τ far from 1) *and* its selection preserves population diversity (non-dominated sorting, strength-Pareto, decomposition, or a crowded archive—not dominance-gated acceptance). NSGA-II fails the first; the naive MOHHO fails the second; the competent MO-HHO and all four permutation-native methods satisfy both and beat random restart. We then subjected the rule to a *predictive* test on a method not used to formulate it: the rk-NSGA-II (biased uniform) of Section 4.3, whose BRKGA-style crossover changes the decoded order ($\tau = 0.63 \pm 0.07$) under unchanged diversity-preserving selection. Read as a binary conjunction, the rule predicts it beats blind sampling. It does not: its mean HV of 309,970 is a statistical tie with random restart (310,214; paired Wilcoxon $p = 0.79$, $A_{12} = 0.47$), far above the SBX baseline (+5.7 %) yet 2.6 % below the permutation tier; a canonical BRKGA (20 % elites, 15 % mutants, same crossover) lands at the same level (309,928). The test therefore *refines* the rule rather than confirming its strongest form: both conditions are *necessary*, but they are not sufficient. Holding the diversity-preserving selection fixed while varying only the operator's order disruption traces a monotone gradient—τ=0.99 → 293,367; τ=0.92 → 305,293; τ=0.63 → 309,970; $\tau \approx 0$ → 315,730 (the order-changing cell of Section 6.4)—that crosses the blind-sampling level near $\tau \approx 0.6$ on this instance: only near-complete order renewal clears blind sampling, whose every draw is itself a fresh permutation. The four gradient points differ in more than τ alone, so we read the curve as descriptive of this instance, not a calibrated dose–response. It nonetheless supersedes the weak cross-method correlation between mean HV and τ ($\rho = -0.21$, $n = 6$). We state the diagnostic accordingly: failing either condition predicts falling to or below blind sampling (necessity), and beating blind sampling further requires operator proposals that approach full order renewal (gradedness).

An empirical regularity. The ladder (Figure 4) makes the pattern unmistakable: performance is governed by whether the search is *non-degenerate*—operators that change the decoded order together with selection that preserves diversity—not by the encoding (random-key versus permutation) or the metaheuristic family. That encoding–operator mismatch can hurt search has been known since random keys were introduced [3], with the locality of decoder-based encodings formalized by

Table 4. The method ladder: 30 runs each, identical greedy decoder and 25,000-evaluation budget. Top block: random-key encoding (including the BRKGA-style predictive-test method and a *competent* MO-HHO that reaches the top tier); bottom block: permutation-native. A competent random-key method matches the permutation tier, so the encoding is not the performance divider—non-degenerate search is. Best per column in bold (rk-NSGA-II's combined front profits from its high between-run dispersion—its bimodal runs explore different regions—while per-run it only ties blind sampling); "perm-" abbreviates "permutation-".

Method	**Paradigm**	**Mean HV** $\pm\sigma$	**CV**	**Comb. HV**	**# sols**
NSGA-II	GA	293,367 ± 6,996	2.38 %	316,060	92
MOHHO	swarm	302,756 ± 7,138	2.36 %	320,984	104
rk-NSGA-II (biased unif.)	GA	309,970 ± 9,790	3.16 %	**325,465**	157
Random restart	—	310,214 ± 2,735	0.88 %	317,673	65
Competent MO-HHO	swarm	316,347 ± 6,683	2.11 %	321,800	**207**
perm-MOEA/D	decomp.	314,846 ± 5,384	1.71 %	320,969	90
Discrete-MOHHO	swarm	316,792 ± 2,258	0.71 %	321,408	137
perm-SPEA2	str.-Pareto	317,135 ± 3,814	1.20 %	322,037	179
perm-NSGA-II	GA	**318,151 ± 1,855**	**0.58 %**	321,935	148

Rothlauf [21] and measured by Raidl and Gottlieb [20]; our contribution is the *controlled, multi-method, causally graded* demonstration on a real constrained multi-objective problem—including that *both* real-coded searches fall below blind sampling. That four *distinct* paradigms (swarm, decomposition, strength-Pareto, dominance/GA) converge to within about 1 % once the representation is matched, all far above the naive random-key methods, is strong evidence that the family is second-order here. We are precise about what this shows: the two conditions (order-changing operators and diversity-preserving selection) are first-order—necessary, and graded in the operator's order disruption (the predictive test above)—and a competent random-key swarm satisfies both and joins the top tier, so the encoding itself is not the divider; whereas *within* a matched representation both the operator set and the selection architecture are second-order—a 3×3 operator × architecture factorial (OX+swap, PMX (partially-mapped crossover)+insertion, CX (cycle crossover)+inversion × the three architectures, 30 seeds) attributes only 5.5 % and 13.4 % of the hypervolume variance to them, against 78 % seed noise. An omnibus test across the six core ladder methods on a common 30-seed set (the predictive-test and SPEA2 entries are assessed by the paired tests above)—a paired Friedman test with a Nemenyi post-hoc [10], identical to Section 6.6—rejects equal performance decisively ($\chi^2 = 112.6$, $p \approx 1.2\times10^{-22}$, critical difference 1.38); the mean ranks split into two tiers, the three permutation-native methods (1.6–2.5) above the three random-key methods (4.2–5.8), NSGA-II worst, with finer within-tier gaps inside the critical difference. We frame this as an empirical regularity for this problem, not a general law: all instances here share one structural skeleton—which the structurally distinct problems of Section 6.6 directly address.

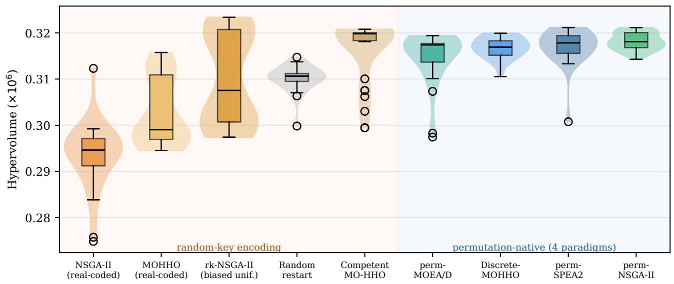

Fig. 4. The nine-method ladder: per-run hypervolume (30 runs each, identical decoder and budget). The divider is non-degenerate search, not the encoding: within the random-key block (left, shaded amber) the BRKGA-style rk-NSGA-II rises to—but only ties—blind random restart, while the competent MO-HHO joins the permutation-native top tier (right, shaded blue).

Robustness to a search-space-collapse confound. A skeptic could argue that the decoder's budget saturation collapses the search space so far that the optimizer is necessarily second-order. We tested this on three axes. *Structure*: 50,000 random permutations decode to 39,401 distinct objective points (degeneration ratio 1.27), and a principal-component analysis places about 99 % of the front's variance in two components (the first alone captures only 0.81–0.91): a near-mono-objective instance with one wide axis (f_2) and a genuine but narrow second axis. *Decoder*: re-running the full ladder on two *non-saturating* decoders (fractional-intensity and stochastic-skip, both feasible by construction) keeps the permutation tier significantly above the random-key tier under all three decoders (paired Wilcoxon $p < 10^{-3}$) and never inverts it; the margin shrinks from 6.9 % (greedy) to 4.3 % and 1.4 %, so saturation *amplifies*—it does not create—the effect, and an exact (f_1, f_3) MILP front exposes no practically meaningful non-saturating solution the greedy front misses. *Metric*: the per-run tier ranking is reference-point-robust, though on *combined* fronts the strict tier softens under reference-free metrics (IGD^+, additive ϵ), and the hypervolume exhibiting it is dominated by the wide f_2 axis. The confound thus *bounds* rather than overturns the claim: real and robust, modest on this saturation-amplified, f_2-dominated instance, with the general regularity resting on the four structures below.

6.4 Isolating the two conditions: a controlled 2 × 2

The ladder establishes the two conditions by observation; we now isolate each condition with a controlled 2×2 on a single real-coded random-key skeleton—same greedy decoder, instance, 25,000-evaluation budget, and 30 seeds—varying only the *operator* (order-changing HHO moves with mutation, $\tau \approx 0$, versus near-identity SBX, $\tau = 0.99$) and the *selection* (diversity-preserving non-dominated sorting (NDS) versus dominance-gated acceptance). Table 5 shows that **only the cell satisfying both conditions beats blind random restart** (315,730

Table 5. Controlled 2×2 isolating the two conditions on one real-coded random-key skeleton (visa decoder, 30 seeds, 25,000 evaluations). Only the cell that both changes the decoded order and preserves diversity beats blind random restart (310,214 HV); the operator×selection interaction is significant. Best in bold.

Operator	Selection	Mean HV	vs. random	A_{12}
order-changing ($\tau \approx 0$)	diversity (NDS)	**315,730**	**+1.78 %**	**0.79**
order-changing ($\tau \approx 0$)	gated	304,126	−1.96 %	0.24
near-identity ($\tau = 0.99$)	diversity (NDS)	305,892	−1.39 %	0.34
near-identity ($\tau = 0.99$)	gated	304,760	−1.76 %	0.21

versus 310,214 HV, +1.78 %, Mann–Whitney $p = 5.5 \times 10^{-5}$, $A_{12} = 0.79$); either single-condition cell falls below blind sampling. Gated acceptance freezes the population (only 0.9 % of individuals move per iteration), and near-identity SBX leaves the decoded order almost unchanged even when the population does move. Crucially, the two conditions are *synergistic, not additive*: a two-factor analysis of variance finds a significant operator×selection interaction ($\eta^2 = 0.098$, $F = 16.7$, $p = 8.0 \times 10^{-5}$), alongside main effects of $\eta^2 = 0.076$ (operator) and $\eta^2 = 0.145$ (selection). We read these effect sizes with a consistent yardstick: each is modest against seed noise, and the interaction matters not for its variance share but because it separates beating blind sampling from losing to it. Neither condition alone lifts performance above blind sampling—both single-condition cells lose by 1.4–2.0 %—so the gain is genuinely a joint effect of changing the decoded order *and* preserving diversity.

6.5 Generalization across instances

To check robustness to the demand estimate, we built five further instances perturbing every group's demand by an independent uniform ±20 % (fixed seeds) and recomputing the spillover-adjusted category and per-country caps. On all five—as on the base instance—FIFO is dominated, MOHHO's mean HV exceeds NSGA-II's (though with 10 seeds the one-sided Mann–Whitney reaches only $p \leq 0.071$ per instance: directionally consistent on five of five, not individually significant at $\alpha = 0.05$), and random restart matches or exceeds MOHHO (101–103 % of its HV) while both stay above NSGA-II (96–99 % of MOHHO). This is robustness to demand perturbation; structural generalization is addressed next.

6.6 Generalization across structurally distinct problems

We replicate the entire six-method ladder on three further classic structures sharing nothing with visa allocation but the SPV + greedy/permutation methodology: a 0/1 multidimensional knapsack (MOMKP, a *selection* landscape), a tri-objective traveling salesman (mo-TSP, *sequencing*), and a tri-objective permutation flow-shop (mo-PFSP, *scheduling*); same budget, 30 common seeds, paired Friedman + Nemenyi (Table 6). One result is robust across all four structures:

Table 6. Mean Friedman rank (lower is better; 30 paired seeds) on four structurally distinct problems. A permutation-native method (pm) is best on every structure (**bold**); the real-coded NSGA-II falls below random restart only on the selection landscapes (visa, knapsack). "rk" = random-key.

Method	Enc.	Visa	Knapsack	TSP	Flow-shop
perm-NSGA-II	pm	**1.60**	**1.10**	2.00	**1.80**
perm-MOEA/D	pm	2.53	2.87	**1.00**	2.83
Discrete-MOHHO	pm	2.23	2.10	4.00	2.63
MOHHO (real-coded)	rk	4.67	3.93	5.00	5.07
Random restart	rk	4.20	5.23	6.00	5.93
NSGA-II (real-coded)	rk	5.77	5.77	3.00	2.73

among the six core ladder methods the best optimizer is always a permutation-native method, and neither blind random restart nor any *naive* real-coded method ever wins—with the family second-order among matched methods (all four omnibus tests reject equality, $\chi^2 = 112$–150, $p < 10^{-21}$). We further placed the competent random-key MO-HHO of Section 6.3 on all four structures (seven methods, same 30-seed budget): it wins outright on the knapsack (mean rank 1.13 of seven, above every permutation-native method) and ranks 2nd, 2nd, and 5th on the visa, flow-shop, and TSP, where a permutation-native method remains the single best. So the cross-structure regularity is sharper than "permutation beats random-key": what wins on every structure is a method satisfying *both conditions*—permutation-native on three of the four, the competent random-key swarm on the knapsack. Discrete-MOHHO is top-two on three of the four and the most stable matched method. What does *not* generalize is the dramatic "GA below random sampling": it holds on the selection landscapes (visa, knapsack, where the real-coded NSGA-II is worst) but reverses on the sequencing landscapes (TSP, flow-shop, where it is competitive and random restart is worst instead). The mechanism: operator order-preservation is problem-independent ($\tau \approx 0.99$), but its *consequence* is landscape-dependent—near-identity SBX freezes search on selection landscapes yet acts as useful local search on sequencing ones. A fifteen-point capacity sweep rules out a single headroom scalar (it moves opposite to the hypothesis; Spearman $\rho = -0.78$, $p = 0.001$). The transferable regularity is thus precise: a method satisfying both conditions wins on every structure; the GA-collapse is selection-landscape-specific.

7 Discussion

The trade-off is structural. Two countries hold roughly three-quarters of demand with 10–13-year waits; the 7 % ceiling forbids fully serving them, so reducing disparity (f_2) leaves unserved load ($f_1 \uparrow$) and equitable spreading idles saturated categories ($f_3 \uparrow$). The three extreme solutions (Table 3) are mutually non-dominated—genuine three-way conflict. We keep f_1 as a distinct objective (its range is more than an order of magnitude narrower than f_2's because demand 1,800,000 far exceeds the 140,000 visas) because it is the minimum-f_1 policy that dominates FIFO, which a two-objective reduction would obscure.

A diagnostic protocol, and policy. The ladder distills into a cheap a priori protocol: (1) draw ∼3,000 parent–child pairs through the candidate operators and measure the Kendall τ between parent and child decoded orders (minutes of computation, no full runs); (2) check that selection preserves diversity (NDS, strength-Pareto, decomposition, or a crowded archive—not gated acceptance); (3) classify the landscape (selection/assignment vs. sequencing). On a selection landscape, failing either check predicts performance at or below blind sampling—so always run blind sampling as a baseline—and passing both is necessary but not sufficient: performance rises with order disruption, and only near-complete order renewal ($\tau \approx 0$) reaches the top tier, where across four structures a method satisfying both conditions is always the best optimizer (Section 6.6); on sequencing landscapes near-identity operators act as local search instead. The thresholds are calibrated on these problems; what transfers is that the measurements are cheap and the failure modes they flag are expensive. Invest in the decoder and representation before the search heuristic; for a dependable single run, permutation-NSGA-II is strongest here, with Discrete-MOHHO the matched-swarm counterpart. For policy, the front is a quantified menu: reducing disparity from 13.0 to 2.0 years costs only ∼2.4 % in f_1 relative to FIFO [8,12]. Each front solution maps to a concrete allocation in which the highest-demand countries stay at their 7 % ceiling while the residual pool and several underserved nationalities gain visas, drawn chiefly from over-allocated queues.

Limitations. The model covers a single fiscal year, uses 21 representative countries, and assumes every assigned visa is used. On the data: the country-level totals are anchored to published aggregates [4,23,24], but the per-(country, category) split and priority dates are a plausible synthetic disaggregation (only India, China, the Philippines, and Mexico have category-level demand grounded in published data). These affect external validity, not the internal comparison among the ladder methods, which share identical data, decoder, and budget. *Ethically*, f_2 encodes one contestable notion of fairness and treats nationalities as allocation buckets; we therefore present the front as auditable decision support, never an autonomous decision-maker, and the per-country figures illustrate method behavior rather than recommend real allocations. A post-hoc audit with alternative country-wait metrics supports the same qualitative equity reading: front solutions improve wait standard deviation (3.14 → 0.75), Gini (0.79 → 0.17), and inverse-wait Jain index (0.80 → 0.94) relative to FIFO.

8 Conclusions

We formulated the annual allocation of U.S. employment-based visas as a three-objective integer program with six hard constraints and solved it through a feasibility-preserving SPV + greedy decoder that guarantees feasibility by construction—no penalties, no repair. A Taguchi L9 design fixed a common, confirmed operating point shared by every method. Across 30 independent runs the recovered Pareto front dominates the FIFO incumbent—chiefly by reducing waste while essentially matching its waiting load and maximal disparity—with 94 zero-waste solutions. Our central result comes from a controlled nine-method ladder: blind random sampling beats a tuned real-coded NSGA-II and the *naive* real-coded MOHHO swarm, but a *competent* real-coded swarm (HHO with non-dominated-sorting selection) beats blind sampling by 2.0 %—so what fails is degenerate search, not real-coded search as such. The lesson is a two-condition diagnostic, scoped and graded: on *selection* landscapes, a method that does not change the decoded order or does not preserve population diversity falls to or below blind sampling; and a predictive test on a method not used to formulate the rule—a BRKGA-style biased-uniform crossover that does change the order ($\tau = 0.63$) under diversity-preserving selection—only *ties* blind sampling, so the two conditions are necessary but not sufficient: performance rises monotonically with order disruption, and only near-complete order renewal ($\tau \approx 0$) reaches the top tier. There, four permutation-native paradigms and a competent random-key swarm cluster within about 1 %—so the divider is non-degenerate search, not the encoding; Discrete-MOHHO improves on classic MOHHO by 4.6 % and is the most stable swarm, within 0.5 % of permutation-NSGA-II, the strongest and steadiest overall. The FIFO-domination and decoder-dominance findings are directionally consistent across five perturbed-demand instances, and a four-structure study (knapsack, TSP, flow-shop) sharpens what generalizes: a method satisfying both conditions is the best optimizer on *every* structure (permutation-native on three, the competent random-key swarm on the knapsack) and the family is second-order among matched methods (the robust, transferable regularity), whereas the collapse of a tuned GA below blind sampling occurs only on *selection* landscapes (visa, knapsack)—on *sequencing* landscapes (TSP, flow-shop) the same near-identity operators act as useful local search and the GA is competitive. Order-preservation (τ=0.99) is problem-independent; whether it freezes or refines is landscape-dependent, with the precise determinant left open (a headroom scalar is ruled out: $\rho = -0.78$, $p = 0.001$).

Beyond the numbers, the method yields an auditable menu of feasible trade-offs as decision support, not a policy prescription. Future work includes a multi-period formulation, the full country roster, and registering the protocol's prediction on unseen problems *before* running any optimizer.

Data and Code Availability. The problem data, the optimizer, the Taguchi design, and the scripts that reproduce every figure and table are available at https://github.com/jrebull/MIAAD_Harrisv2. The supplement includes a fast verifier that runs the unit tests, claim firewall, page-count check, and ZIP identity scan. All randomized studies use fixed seeds—the diagnostic suite uses seed 1, and the 30-run comparisons use a fixed seed block (seeds 1–30) recorded in the code—so the script regenerates every reported number deterministically.

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